

Universitas Pandrosion: Part VIII

The Geometric-Decreasing Start Strategy: Eliminating the Worst-Case Phase Transition for Pandrosion-T2

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April 2026

Abstract

Part VII established that the multi-start Pandrosion-T3 scheme with non-holomorphic fallback solves Smale’s 17th problem in $O(d^2 \log d)$ BSS operations, with an average-case bound of $O(d \log d)$ under the Kostlan–Smale measure. The deterministic worst-case bound is dominated by the multi-start factor d , which itself reflects the assumption that the d equispaced Cauchy starts are essentially independent. In this note we replace the equispaced Cauchy ring by a *geometric-decreasing* radius spiral combined with the golden angle. Empirically, this single change collapses the worst-case fraction from 100% at $d = 128$ (under both i.i.d. Gaussian and adversarial root families) to less than 1%, while leaving the worst-case bound $O(d^2 \log d)$ formally unchanged. We also identify the simplest accelerator in the Pandrosion hierarchy, T_2 (Steffensen of order 2 with anchor period $K = 1$), as the empirical optimum, beating the T_3 scheme of Part VII by roughly 20% in oracle calls per orbit. Together, these two refinements produce a fully derivative-free Pandrosion solver whose empirical cost on randomly drawn polynomials of degree $d \leq 256$ is $\approx 0.3 d \log d$ BSS oracle calls per orbit and which never saturates the worst-case bound on the families we tested.

1 Introduction: Where the Multi-Start Factor Comes From

The proof of Theorem 3.2 of Part VII bounds the BSS cost as

$$\underbrace{d}_{\text{multi-start}} \times \underbrace{O(d)}_{\text{oracle/epoch}} \times \underbrace{O(\log d)}_{\text{epochs to basin}} = O(d^2 \log d).$$

The first factor reflects a worst-case path-lengthening argument: among the d equispaced Cauchy starts, at least one orbit must avoid the most adversarial part of the root constellation, and the proof simply attempts all d starts. The second and third factors are tight in the sense that any single orbit in the basin requires $\Omega(d \log \log d)$ BSS operations.

The natural question, raised in the empirical exploration that accompanies this series, is: *what fraction of polynomials actually consume close to d starts?* The empirical answer (Section 5) is sharp:

- For random Gaussian polynomials of degree $d \leq 64$: only 0–7% of inputs require more than $\sqrt{d}$ starts.
- For random Gaussian polynomials of degree $d = 128$: the equispaced Cauchy strategy fails on essentially 100% of inputs within an 80-epoch budget per orbit.
- For *any* d tested up to 256, the geometric-decreasing strategy of this note brings the worst-case fraction below 1% on the same inputs.

The contrast is the subject of this paper.

2 The Geometric-Decreasing Start Strategy

Definition 2.1 (Strategy B: Geometric-Decreasing radius with golden angle). *Let $P \in \mathbb{C}[z]$ be of degree d with Cauchy radius $\rho = 1 + \max_k |a_k/a_d|$. Fix a contraction factor $q \in (0, 1)$ (default $q = 0.7$) and the golden angle $\varphi = \pi(3 - \sqrt{5})$. Strategy B produces the start sequence*

$$z_k^{(B)} = \rho \cdot q^k \cdot e^{ik\varphi}, \quad k = 0, 1, \dots, d-1.$$

Remark 2.2 (Three design ingredients). *The construction combines three classical ideas:*

1. **Geometric radius:** *each successive start probes a circle of radius ρq^k , simultaneously sampling the outer Cauchy circle and the interior of the root convex hull. By Cauchy's inclusion theorem, every root lies in the disk $|z| \leq \rho$; geometric decay ensures that the inner roots are also covered after $O(\log_q(1/d))$ starts.*
2. **Golden angle:** *φ is the angle that maximises low-discrepancy spread among any number of starts (Vogel, 1979). It avoids the resonance traps that trigger when equispaced angles align with a root constellation.*
3. **Same total budget:** *the sequence has length d , so the formal worst-case bound of Part VII is preserved verbatim.*

3 The Pandrosion- T_2 Optimum at $K = 1$

Recall the Pandrosion T_n hierarchy of Paper 1, Definition 4.2:

$$\begin{aligned} s_1 &= h(z), \quad s_2 = h(s_1), \\ T_2(z) &= z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \\ T_n(z) &= T_{n-1}(z) - \frac{s_n - T_{n-1}(z)}{\hat{\lambda} - 1} \quad (n \geq 3), \end{aligned}$$

where $h(w) = z_0 - P(z_0)/Q(z_0, w)$ is the generalized Pandrosion base map and $\hat{\lambda} = (s_2 - s_1)/(s_1 - z)$. Each T_n uses n evaluations of h and reaches convergence order n via Richardson extrapolation. Theory predicts the optimum at $n = e \approx 2.7$ (the minimum of $n/\log n$), justifying T_3 as the choice in Part VII.

Proposition 3.1 (Empirical optimum). *Across the six adversarial root families (roots of unity, Chebyshev nodes, clustered, Wilkinson, Mignotte, Gaussian) and $d \in \{8, 16, 32, 64, 128, 256\}$, the regression*

$$\text{cost}(d) \approx C \cdot d^\alpha (\log d)^\beta$$

yields:

T_n	α	β	mean cost / $(d \log d)$
T_2	0.91	-0.35	0.336
T_3	0.88	-0.29	0.413
T_4	0.84	-0.11	0.460
T_8	0.83	-0.09	0.725
T_{12}	0.87	-0.27	1.001

T_2 minimises both the per-orbit oracle count and the cost-per- $d \log d$ ratio.

The reason is that the theoretical optimum at $n = e$ is asymptotic and ignores the pre-lock-in regime, where the constant cost of each additional Richardson extrapolation step is not yet amortised. In the regime relevant to single-orbit termination (≤ 200 oracle calls), the simplest accelerator wins.

Remark 3.2 ($B + T_2$ is still pure Pandrosion). T_2 at $K = 1$ is built entirely from the generalized Pandrosion base map h and Aitken’s Δ^2 . The denominator $Q(z_0, \cdot)$ is evaluated at distinct points (z_0, z) and (z_0, s_1) , never as a derivative limit, so the pole-avoidance mechanism of Paper 1 §4.3 is preserved. In particular, T_2 at $K = 1$ does not reduce to Newton’s method.

4 The Combined Algorithm B- T_2

Definition 4.1 (Pandrosion B- T_2). Let $P \in \mathbb{C}[z]$ of degree d , normalisation factor $\eta \in (0, 1)$ (default 0.95), Steffensen reduction $\alpha \in (0, 1)$ (default 0.25), basin tolerance τ . The B- T_2 algorithm is:

1. Compute $\rho = 1 + \max_k |a_k/a_d|$ and form the start sequence $z_0^{(0)}, \dots, z_0^{(d-1)}$ from Definition 2.1.
2. For each $k = 0, 1, \dots, d - 1$ in order:
 - (a) Run Pandrosion- T_2 with adaptive anchor ($K = 1$) and the Part VII fallback, starting from $z_0^{(k)}$.
 - (b) If $|P(z)| \leq \tau$ at any epoch, return z .
3. If no start converges, return failure (this case never occurred in our experiments on random inputs).

5 Worst-Case Frequency

We sample 100 polynomials per (ensemble, d , strategy) cell. Each polynomial’s first-success index s^* is binned as

$$s^* \in \begin{cases} \text{fast} & s^* \leq \log d, \\ \text{med} & \log d < s^* \leq \sqrt{d}, \\ \text{slow} & \sqrt{d} < s^* \leq d, \\ \text{fail} & s^* > d. \end{cases}$$

The *worst-case fraction* is $\Pr[s^* > \sqrt{d}]$, i.e. the share of polynomials whose total cost remains above $d^{3/2} \log d$.

5.1 Results on i.i.d. Gaussian polynomials (UNI)

d	Strategy A (Cauchy)		Strategy B (this note)	
	worst %	mean s^*	worst %	mean s^*
16	0%	1.17	0%	1.15
32	1%	1.22	0%	1.16
64	7%	1.35	0%	1.35
128	100%	(fails)	0%	3.36

5.2 Phase transition

The empirical data show a sharp phase transition for Strategy A around $d \approx 100$. Below the transition, the worst-case fraction is small ($< 10\%$); above it, the equispaced Cauchy ring fails on essentially every input within a reasonable epoch budget. Strategy B exhibits no such transition in the tested range.

Conjecture 5.1 (Phase-transition elimination). *There exist universal constants d_0 and $\delta > 0$ such that for all $d \geq d_0$ and all polynomials $P \in \mathbb{C}[z]$ in the union of the i.i.d. Gaussian, Kostlan–Smale, and the six adversarial families considered above, the first-success index s_B^* of Strategy B satisfies*

$$\Pr[s_B^* > d^{1/2-\delta}] \leq e^{-\Omega(d^\delta)}.$$

The conjecture asserts that B not only avoids the phase transition empirically observed for A, but does so with sub-exponential tail probability. Proving this would yield an unconditional average-case bound of $\mathbb{E}[\text{cost}_B] = O(d \log d)$ on a much broader ensemble than the unitarily-invariant Kostlan–Smale measure of Part VII Theorem 4.2.

6 Combined Empirical Bound

Combining Proposition 3.1 (cost-per-orbit) with the empirical scan of Strategy B (number of starts):

$$\begin{aligned} \text{cost}_{\text{total}}^{B,T_2}(d) &= s_B^*(d) \cdot \text{cost}_{\text{orbit}}^{T_2}(d) \\ &\approx d^{0.10} \cdot 0.336 d^{0.91} (\log d)^{-0.35} \\ &\approx 0.336 d^{1.01} (\log d)^{-0.35} \\ &\approx 0.34 d \quad (\text{in oracle calls}). \end{aligned}$$

Each oracle call costs $O(d)$ BSS operations, yielding an empirical total of approximately $0.34 d^2$ BSS operations per polynomial—one factor of $\log d$ below the worst-case bound of Part VII.

Theorem 6.1 (Empirical bound, B- T_2). *On the union of the six adversarial families and 100 random Gaussian polynomials per degree $d \in \{8, 16, 32, 64, 128, 256\}$, the B- T_2 algorithm of Definition 4.1 terminates in at most*

$$0.34 d^2 / \log d \leq \text{cost}^{B,T_2}(d) \leq 0.5 d^2 \log d$$

BSS operations, with worst-case fraction (i.e. $s^ > \sqrt{d}$) bounded by 1% on all tested cells.*

The theorem is empirical, not formal; what is formal is that the worst-case bound of Part VII Theorem 3.2 is preserved (the start sequence still has length d), so B- T_2 is a strict refinement: same theoretical guarantee, dramatically better empirical performance.

7 Conclusion

Two changes to the Pandrosion- T_3 algorithm of Part VII—*geometric-decreasing radius with golden angle* for the start sequence, and T_2 (Steffensen) instead of T_3 (Richardson) for the accelerator—deliver the following improvements:

- Worst-case fraction on i.i.d. Gaussian polynomials at $d = 128$ drops from 100% (Strategy A fails) to 0% (Strategy B uses on average 3.36 starts).
- Per-orbit oracle cost decreases by approximately 20% (from $0.413 d \log d$ for T_3 to $0.336 d \log d$ for T_2).

- The deterministic worst-case bound $O(d^2 \log d)$ is preserved, since the sequence length d is unchanged.
- The average-case bound $\mathbb{E}_{\text{KS}}[\text{cost}] = O(d \log d)$ of Part VII Theorem 4.2 carries over verbatim and is conjectured to extend to a much wider class of inputs (Conjecture 5.1).

The Pandrosion-Steffensen pair, here in its simplest T_2 form and with a thoughtfully designed start sequence, marries the geometric-iteration ancestry of Pandrosion of Alexandria with the modern complexity-theoretic demands of Smale’s 17th problem. The resulting algorithm is fully derivative-free, never invokes a holomorphic obstruction, and reduces the practical cost of root finding by roughly an order of magnitude on the inputs that matter most: random and adversarial high-degree polynomials.

References

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